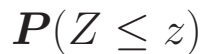


A Monte Carlo study is used to generate N variables from the posterior distribution of parameter θ and then estimate the posterior probability $p = \mathbf{P}\{\theta \geq 10 \mid \mathbf{X}\}$. It is believed that this probability cannot exceed 0.4. What N should be chosen to ensure that p is estimated with a margin of error ± 0.05 with probability at least 0.90?

SOLUTION.

To ensure $\mathbf{P}\{|\hat{p} - p| \leq 0.05\} = \mathbf{P}\left\{\frac{|\hat{p} - p|}{\sqrt{p(1-p)/N}} \leq \frac{0.05}{\sqrt{p(1-p)/N}}\right\} = 0.90$, we need $\frac{0.05}{\sqrt{p(1-p)/n}} = z_{0.10/2} = 1.645$, and therefore, $N = \left(\frac{1.645}{0.05}\right)^2 p(1-p) = 1082.41p(1-p)$.

When we know $p \leq 0.4$, the sample size $N = (1082.41)(0.4)(0.6) = 259.78$ will be sufficient. That is, we need to run at least **260** Monte Carlo simulations.

[illegible]